

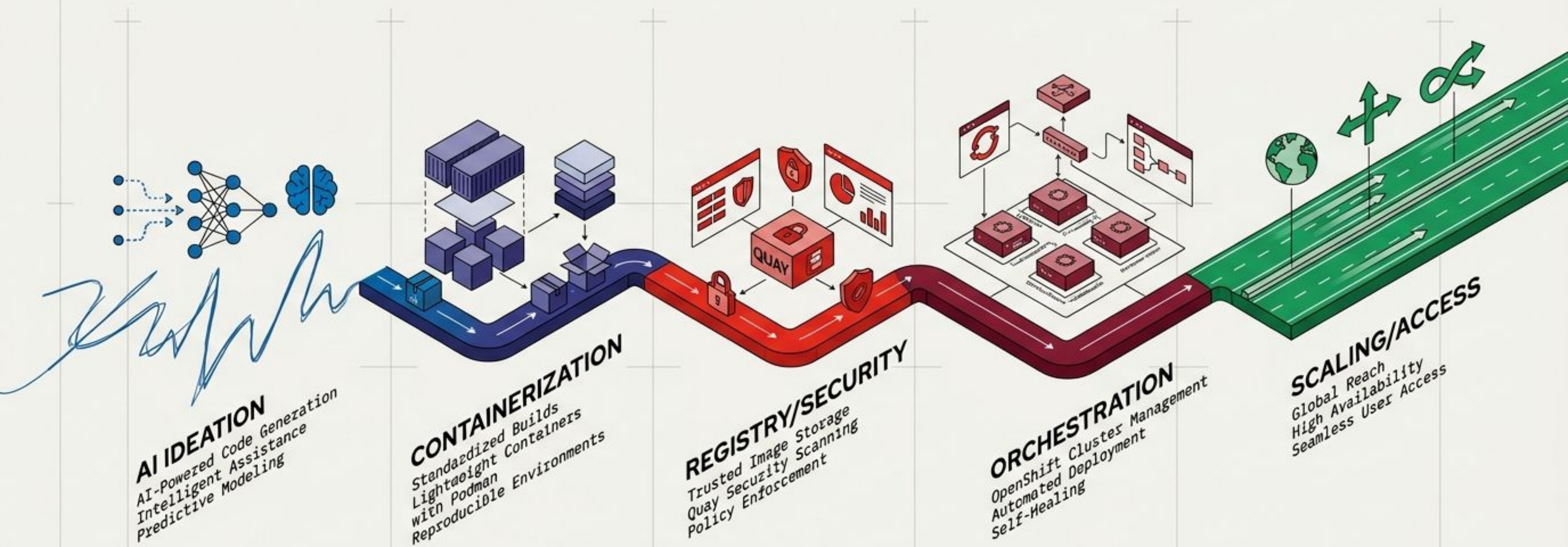
# The Blueprint for modern delivery

Alfred Bach

Principal Instructor

# THE BLUEPRINT FOR MODERN DELIVERY

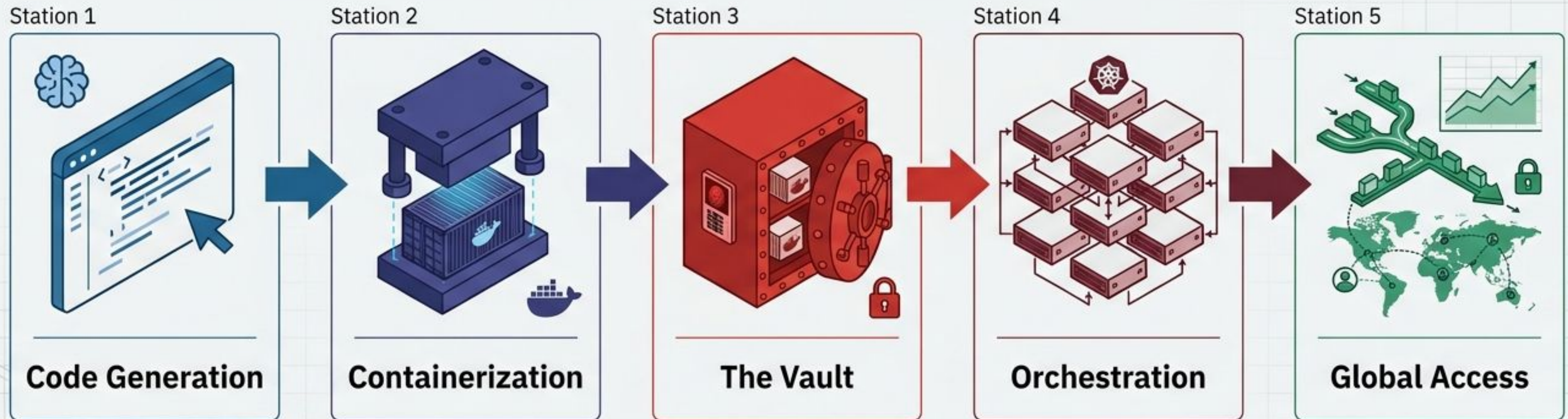
Engineering the code-to-cloud pipeline with AI, Podman, Quay, and OpenShift.





# The Journey: Idea to Global Scale

Raw code is fragile. To survive production, an application requires an automated, secure, and governed pipeline. This is the exact path modern enterprises use to turn ideas into scalable architecture.

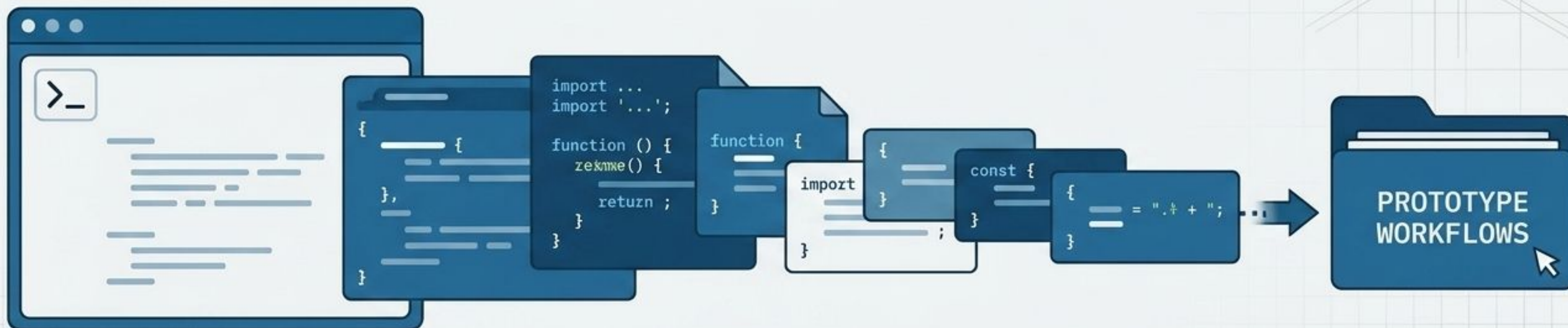


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# Phase 1: Overcoming the Blank Page

AI coding assistants accelerate development from prototype thinking to functional workflows in seconds.



## The “Works on My Machine” Problem

Missing dependencies and environment drift prevent raw code from surviving in production.

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# The Physics of Deployment: Why Containers?

Containers share the host OS kernel, allowing hundreds of instances to run on a single machine. They provide the perfect, immutable mini-environment for AI-generated artifacts to travel seamlessly.

|                | Virtual Machines      | Containers                  |
|----------------|-----------------------|-----------------------------|
| Architecture   | Heavy (Full Guest OS) | Lightweight (Shared Kernel) |
| Startup Time   | Minutes               | Seconds                     |
| Resource Usage | High overhead         | Extremely efficient         |
| Isolation      | Full hardware-level   | Process-level               |
| Portability    | Limited               | Write once, run anywhere    |

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# Phase 2: Forging the Artifact with Podman



## 1. Package

Using a Dockerfile to bundle code, libraries, and runtime.

## 2. Build

Running daemonless builds locally without requiring root privileges.

## 3. Standardize

Complete consistency guaranteed from laptop to cloud.

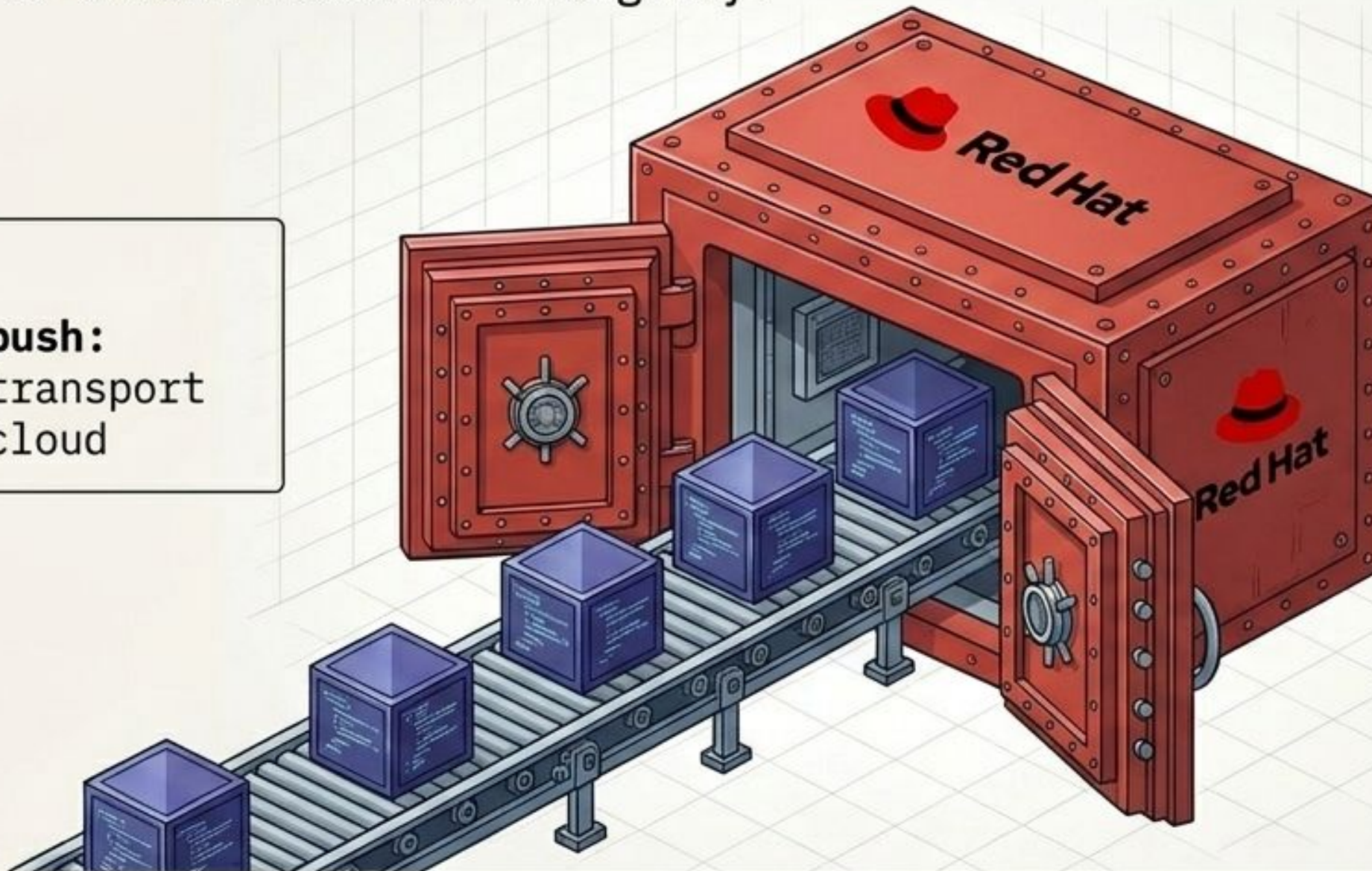


# Phase 3: The Enterprise Vault

Once built, the immutable container must be distributed. Red Hat Quay serves as the centralized, highly available storage repository, utilizing immutable SHA-256 digests to ensure absolute integrity.



**podman push:**  
Secure transport  
to the cloud



**Geo-replication:**  
Enterprise-scale  
distribution



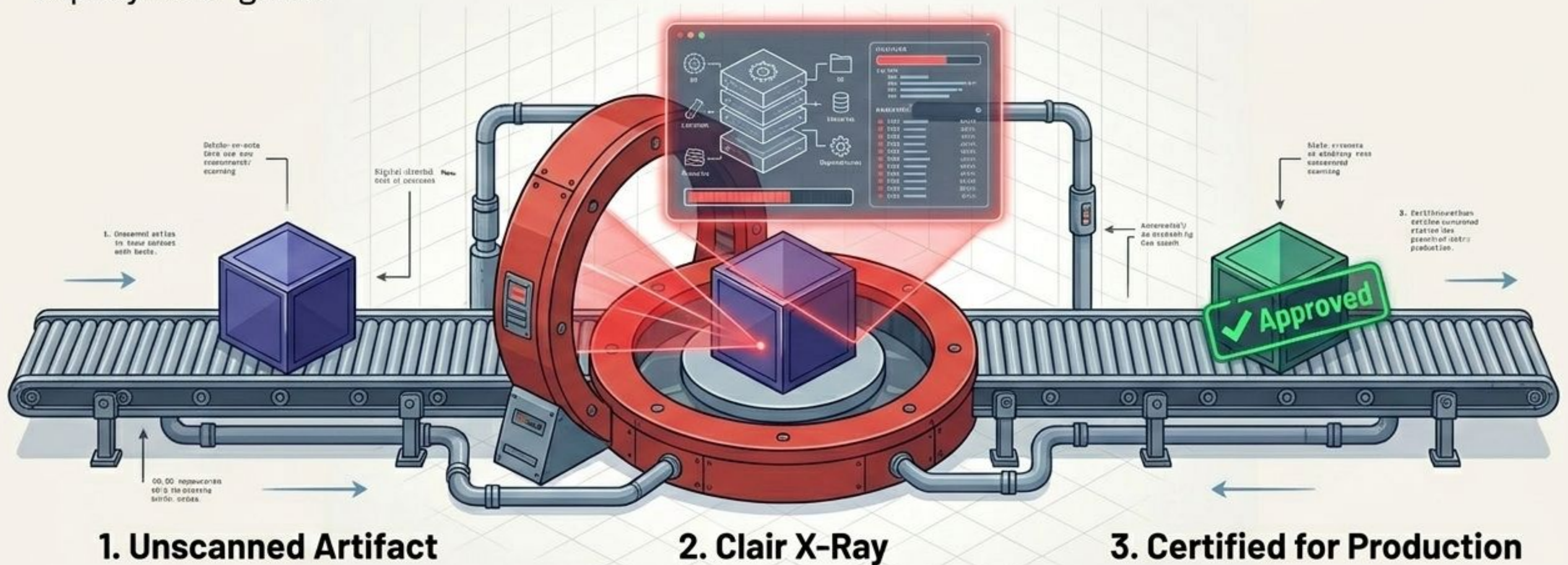
**Granular Access:**  
Strict RBAC  
controls





# The Gatekeeper: Automated Security at Rest

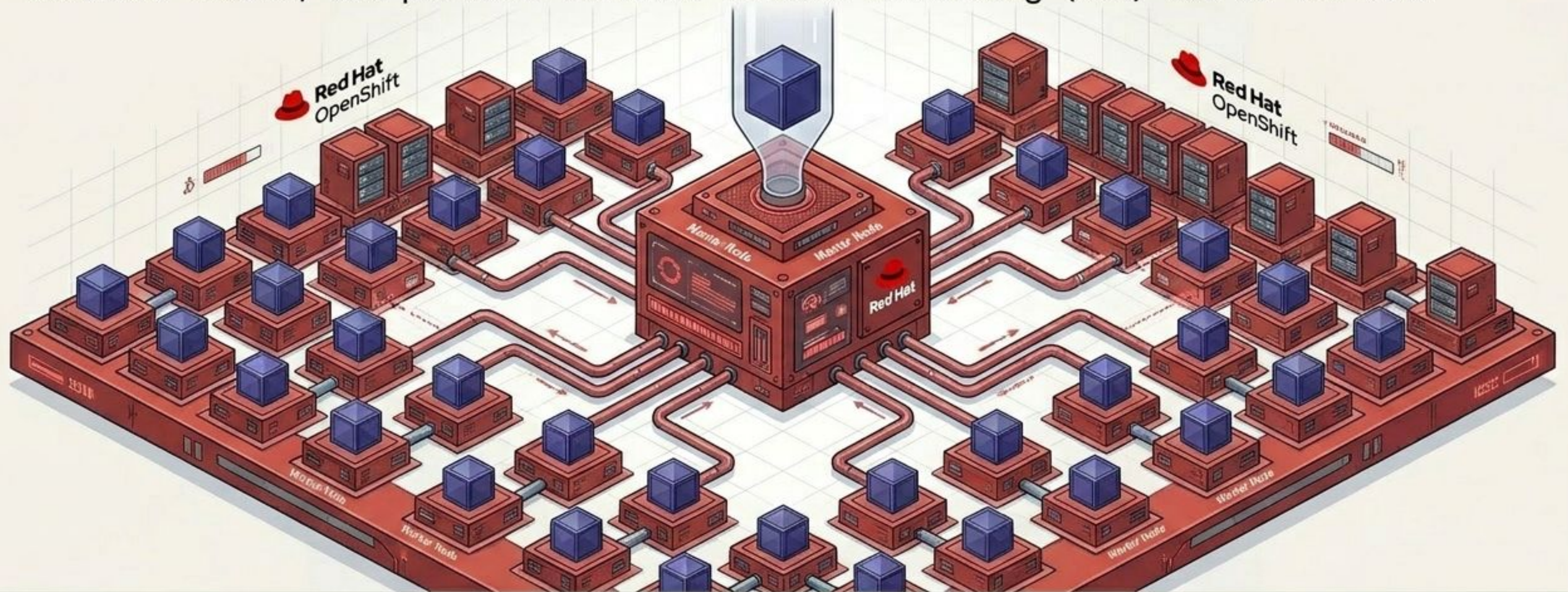
Red Hat Quay integrates Clair vulnerability scanning. It automatically inspects container layers for known CVEs, preventing compromised code from reaching the deployment grid.





# Phase 4: Orchestrating the Grid

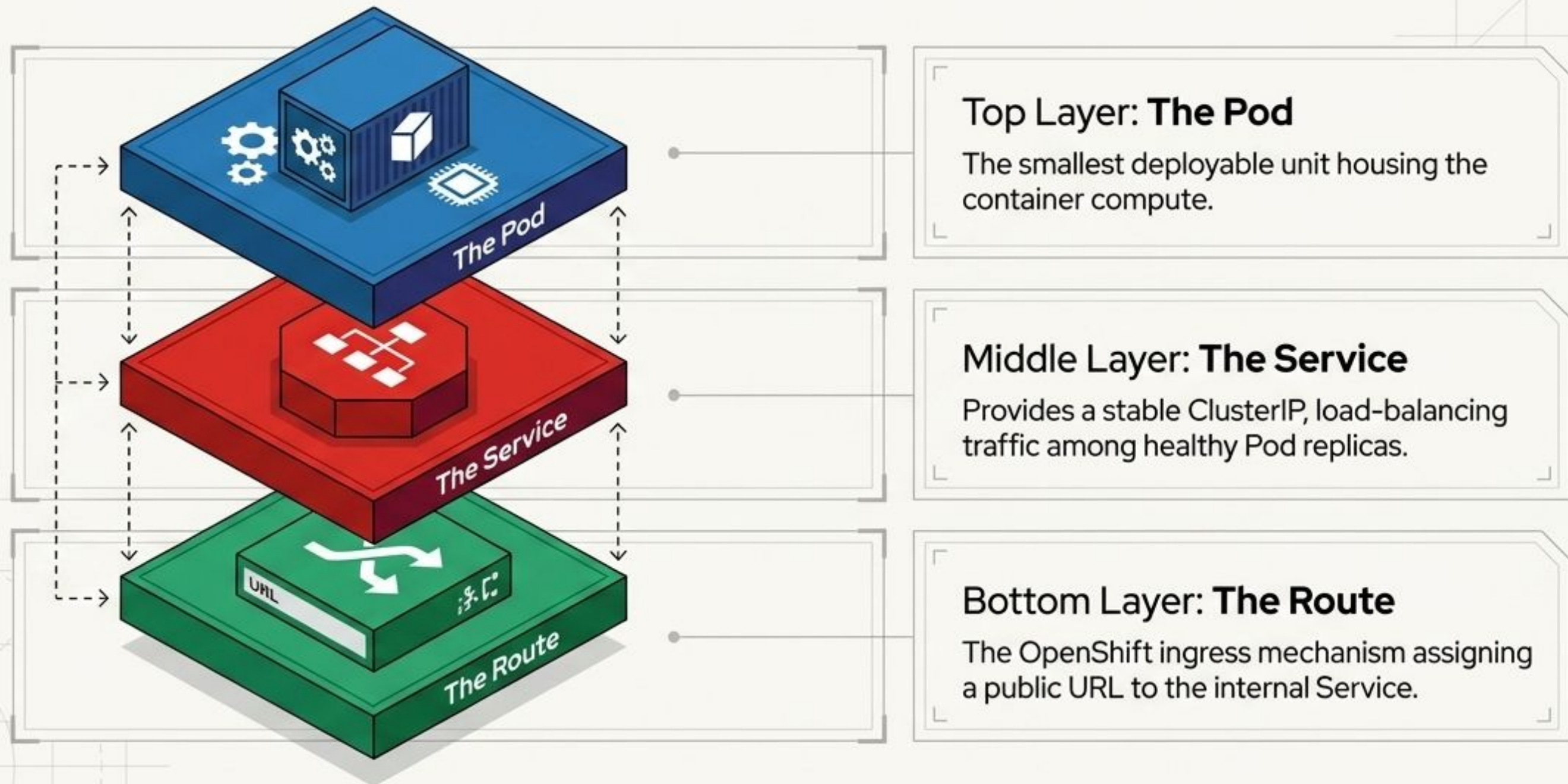
Running one container is easy. Load-balancing, self-healing, and networking thousands requires an orchestrator. Built on Kubernetes, Red Hat OpenShift automates deployment, monitors health, and provides software-defined networking (SDN) out-of-the-box.





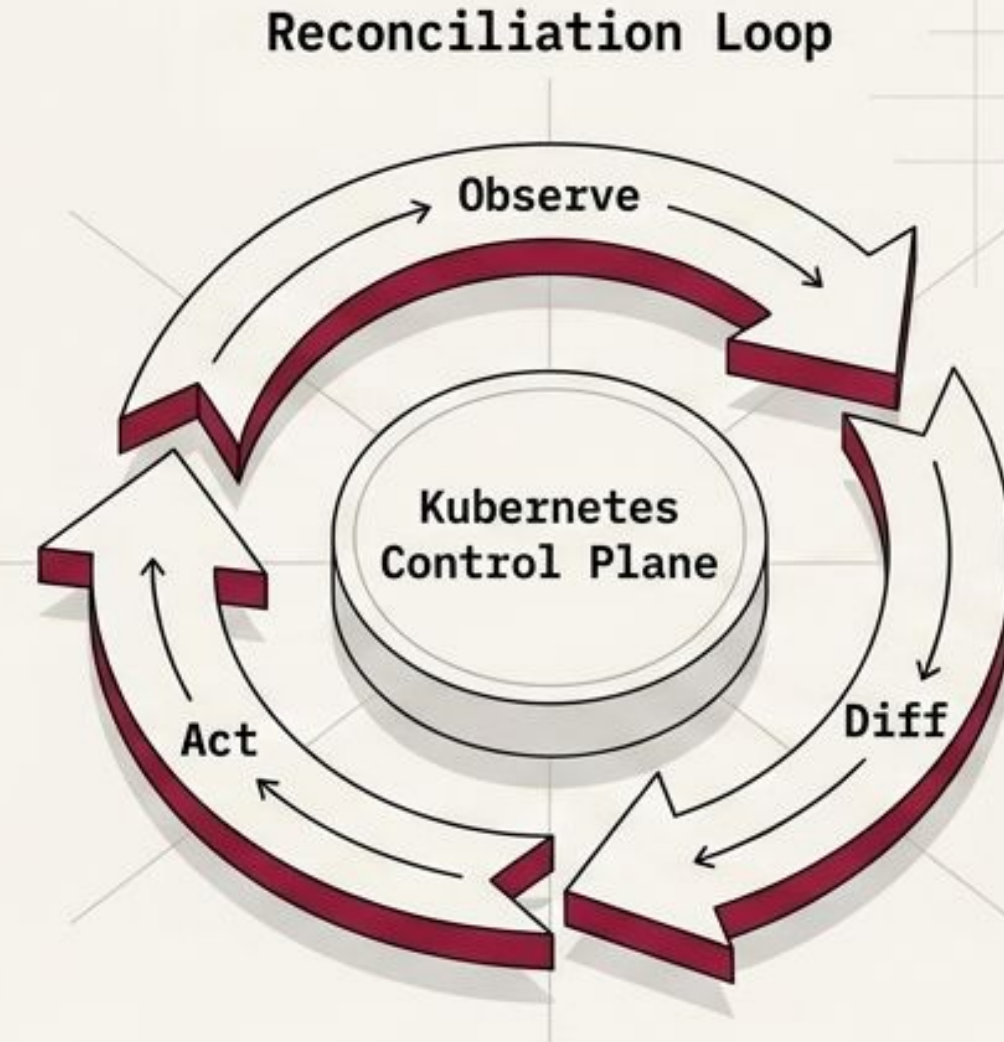
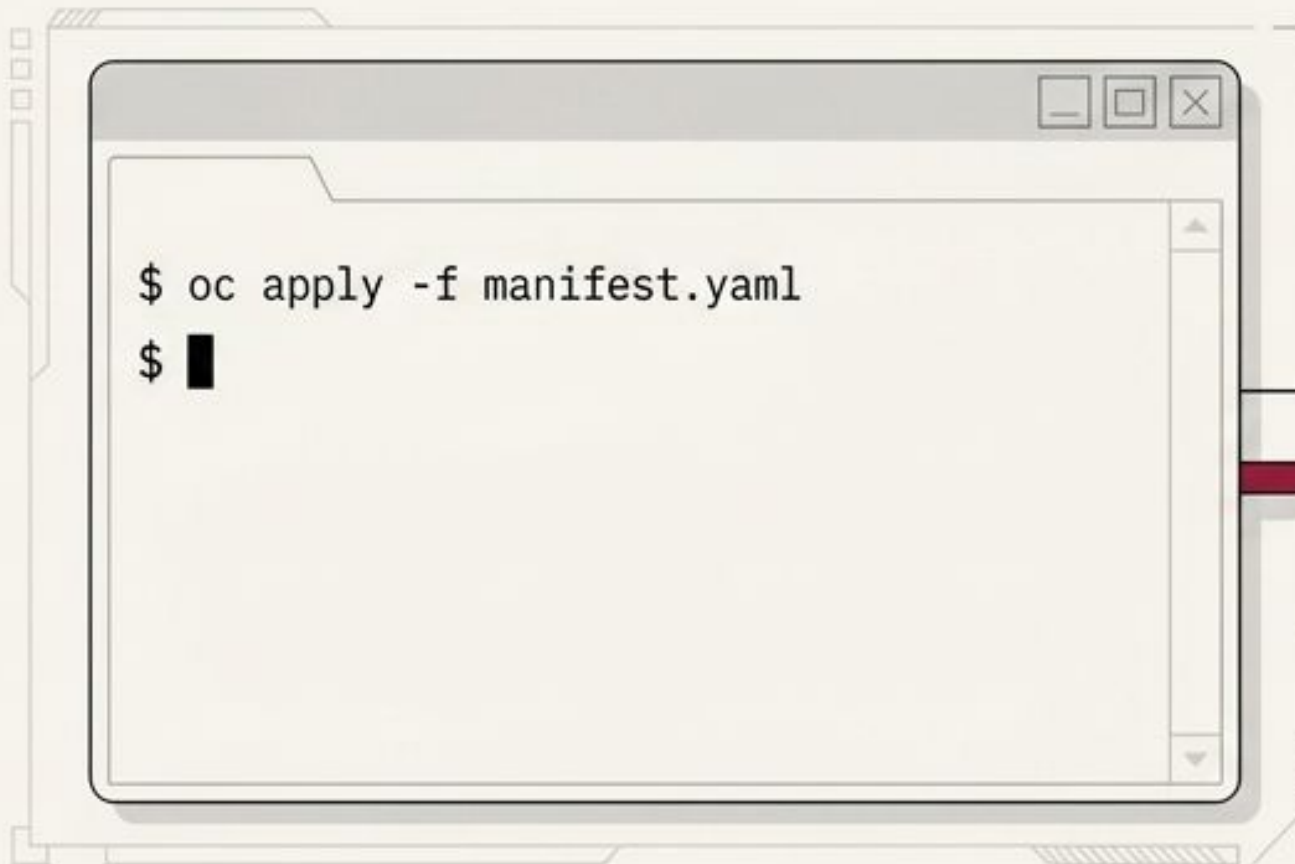
# The Anatomy of an Application

Translating abstract declarative YAML manifests into physical infrastructure.





# State Engine: Commanding the Cluster



**1. Declare:** Developer defines the desired state in a YAML manifest.

**2. Execute:** The 'oc apply' CLI sends the blueprint to the API.

**3. Reconcile:** The OpenShift Control Plane continuously compares actual vs. desired state, automatically self-healing failed pods.





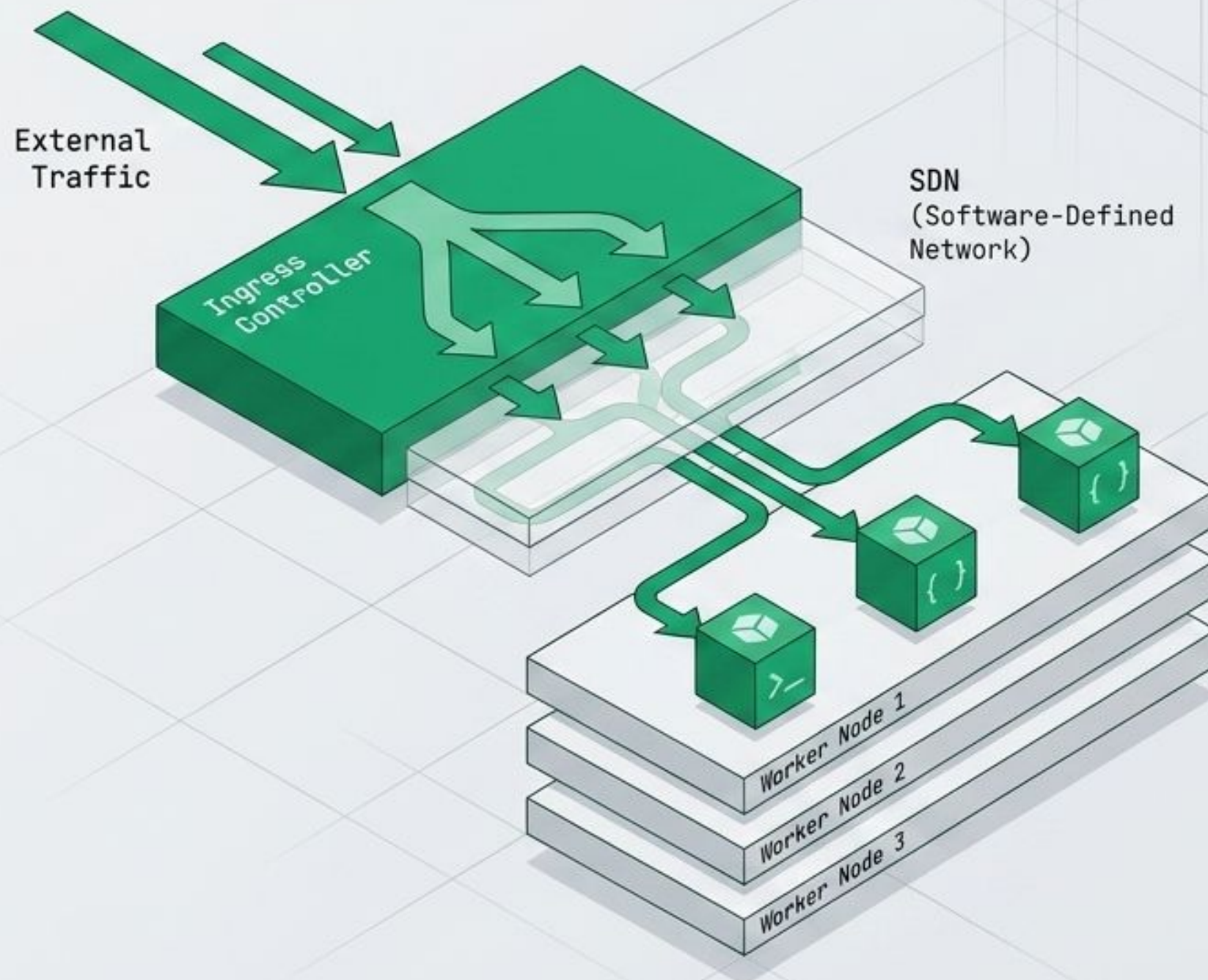
# Phase 5: Routing Global Traffic

## Routes & Ingress

Directs external HTTP/HTTPS traffic to internal services using clean URLs.

## Load Balancers

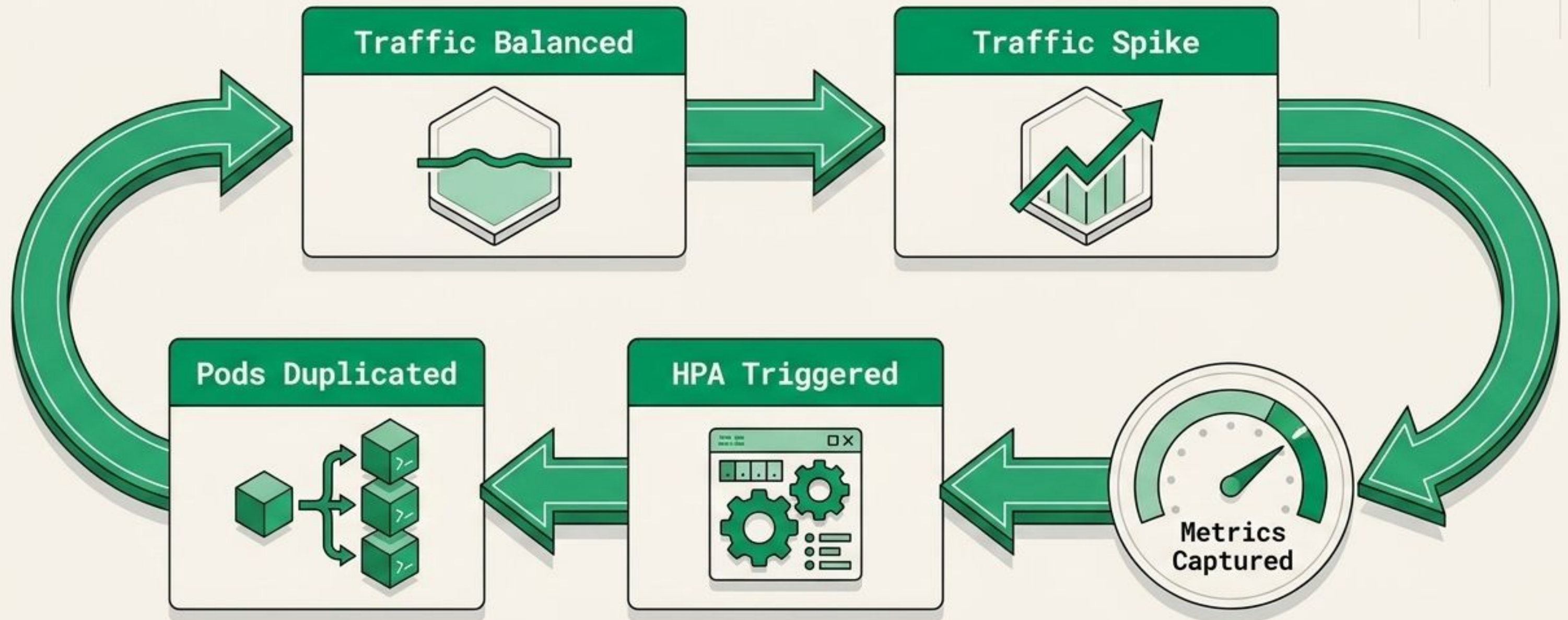
Automatically provisioned to distribute massive inbound connections.





# The Scaling Feedback Loop

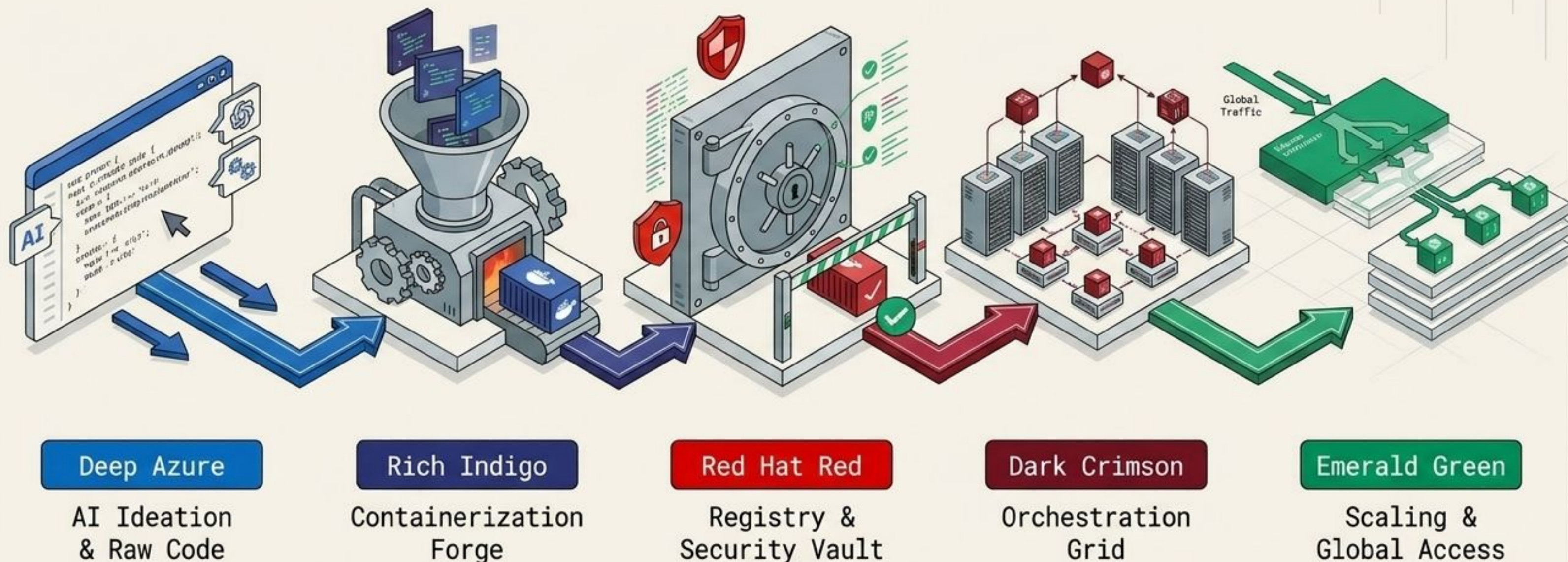
When pods consume all available node capacity, OpenShift automatically provisions entirely new infrastructure nodes to house the overflow.





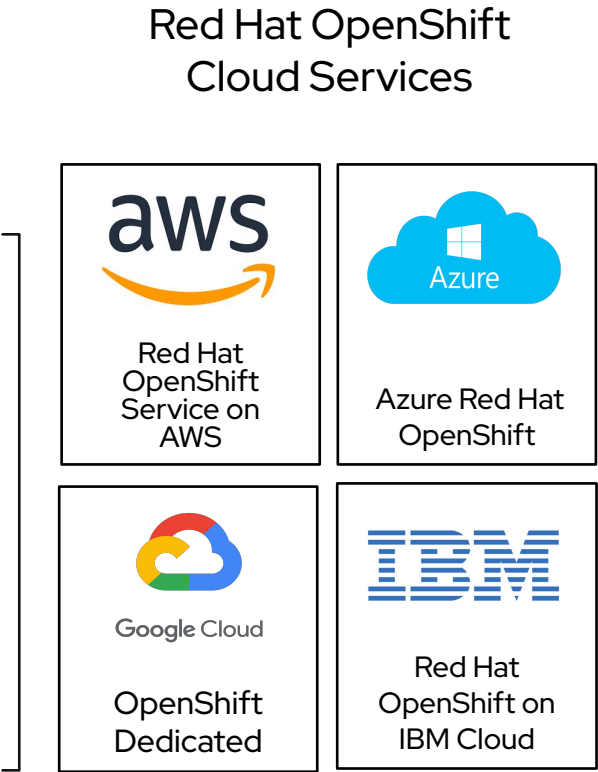
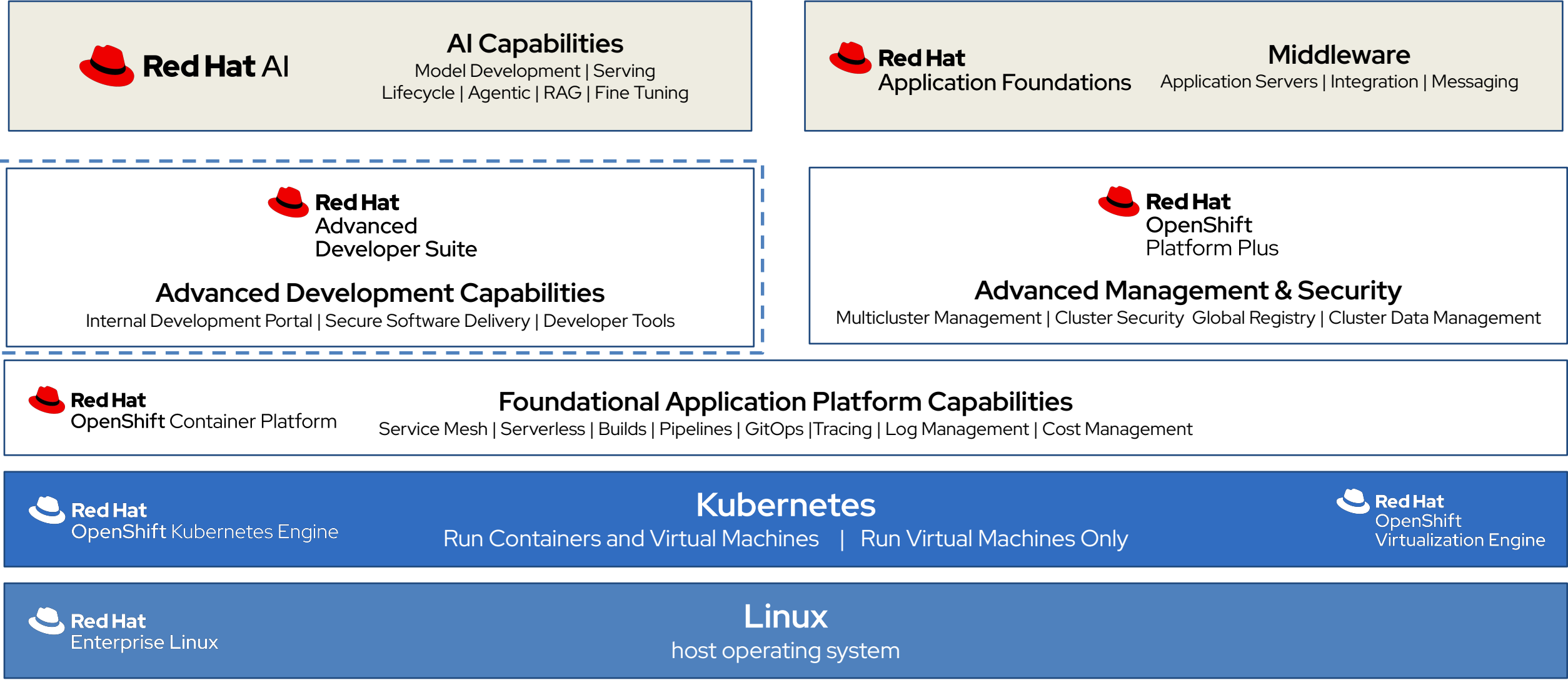
# The Continuous Ecosystem: Prompt to Production

What begins as a fragile line of AI-generated code is transformed into a secure, immutable, self-healing, and infinitely scalable enterprise architecture in minutes.





# Red Hat's Hybrid Application Platform





# Move Fast, Stay Safe

## Red Hat Advanced Developer Suite



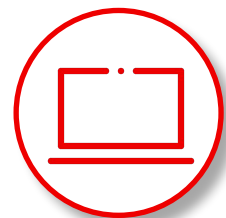
### Enable development teams to deliver business value faster

Create an integrated, self-serve developer experience from end to end with **Red Hat Developer Hub**



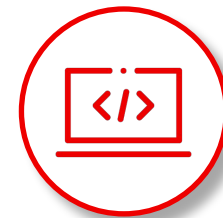
### Reduce supply chain security risks across the software development lifecycle

Trust, transparency across devops processes with **Red Hat Trusted Software Supply Chain**



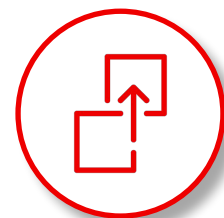
### Extend OpenShift experience to developer desktops

Containers and Kubernetes on the desktop with **Podman Desktop** and **OpenShift Local**



### Standardize Developer tools

Enable application coding locally on the desktop with **IDE Plugins** and remotely in the browser with **OpenShift DevSpaces**

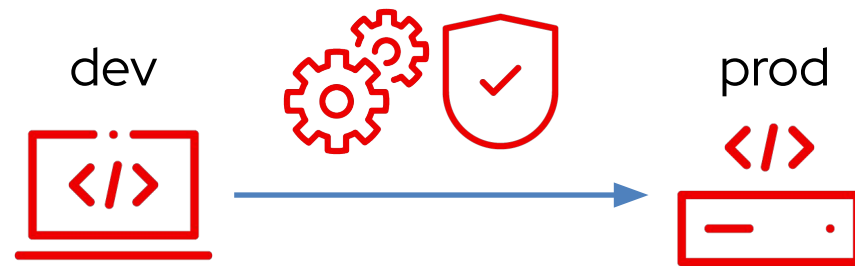


### AI-Accelerated workload migration to OpenShift

Accelerates large scale migrations and modernization with **Migration Toolkits**

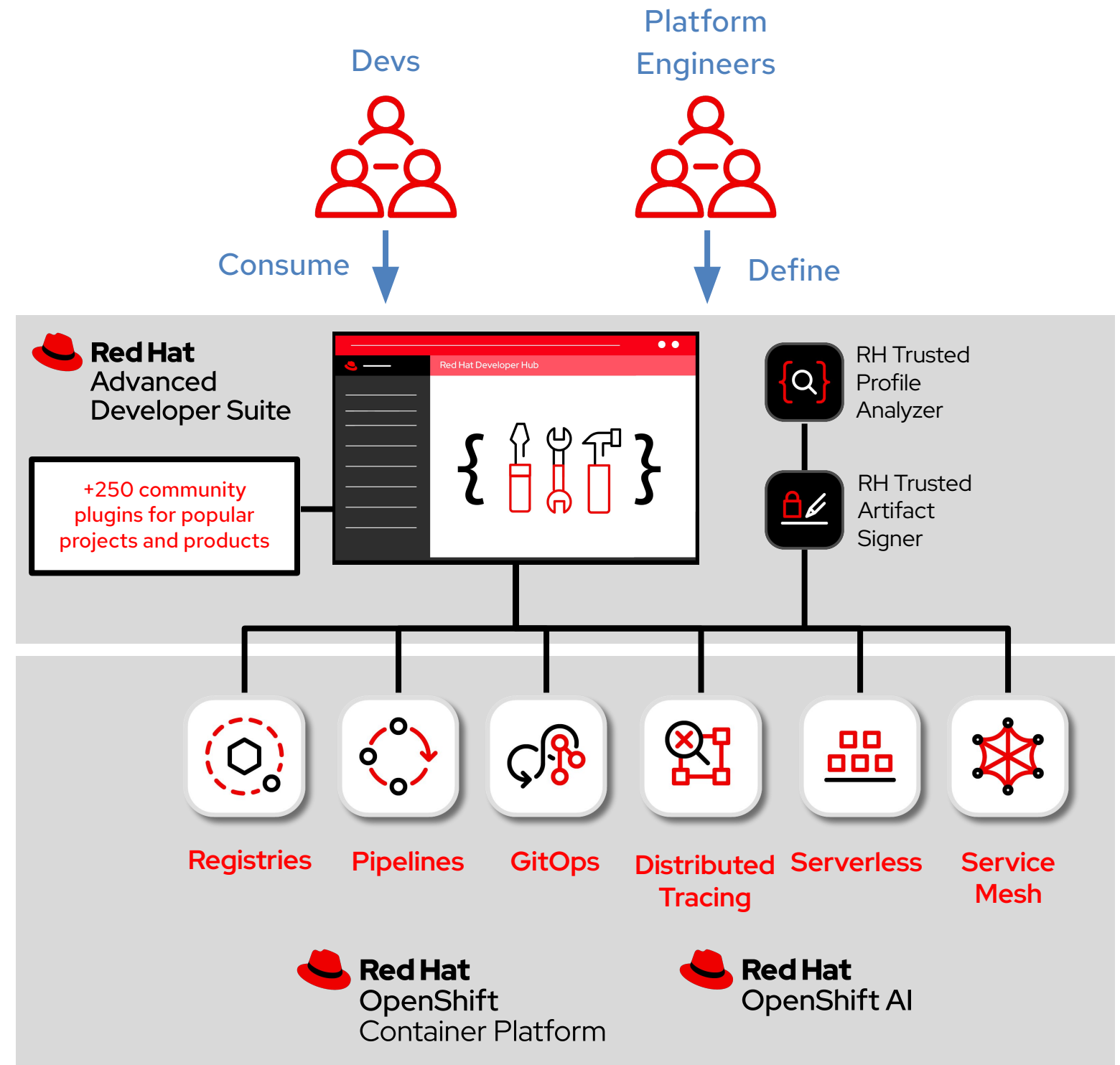


# Red Hat Advanced Developer Suite



RHADS combines platform engineering tools with enhanced security capabilities to help accelerate and simplify application development with new enhancements to speed the adoption of AI

- Modular design allows it to be integrated with existing platforms and systems including Red Hat's platforms
- Integrates with existing security capabilities found in Red Hat Advanced Cluster Security
- Complements existing developer tools from Red Hat including OpenShift DevSpaces
- Built on leading open source projects like Backstage and Sigstore





### INCLUDED PRODUCTS



IDP: Internal  
Developer Portal



**Red Hat**  
Developer Hub



Trusted Software  
Supply Chain



**Red Hat**  
Trusted  
Artifact Signer



**Red Hat**  
Trusted Profile  
Analyzer

### COMPLEMENTARY TECHNOLOGIES



AI/Kube Developer  
Desktop Experience :  
**Podman Desktop**



AI Enablement



Developer Lightspeed



Podman Desktop AI  
Lab

Red Hat Developer  
Hub AI Integration



Development  
Tools  
IDE Plugins

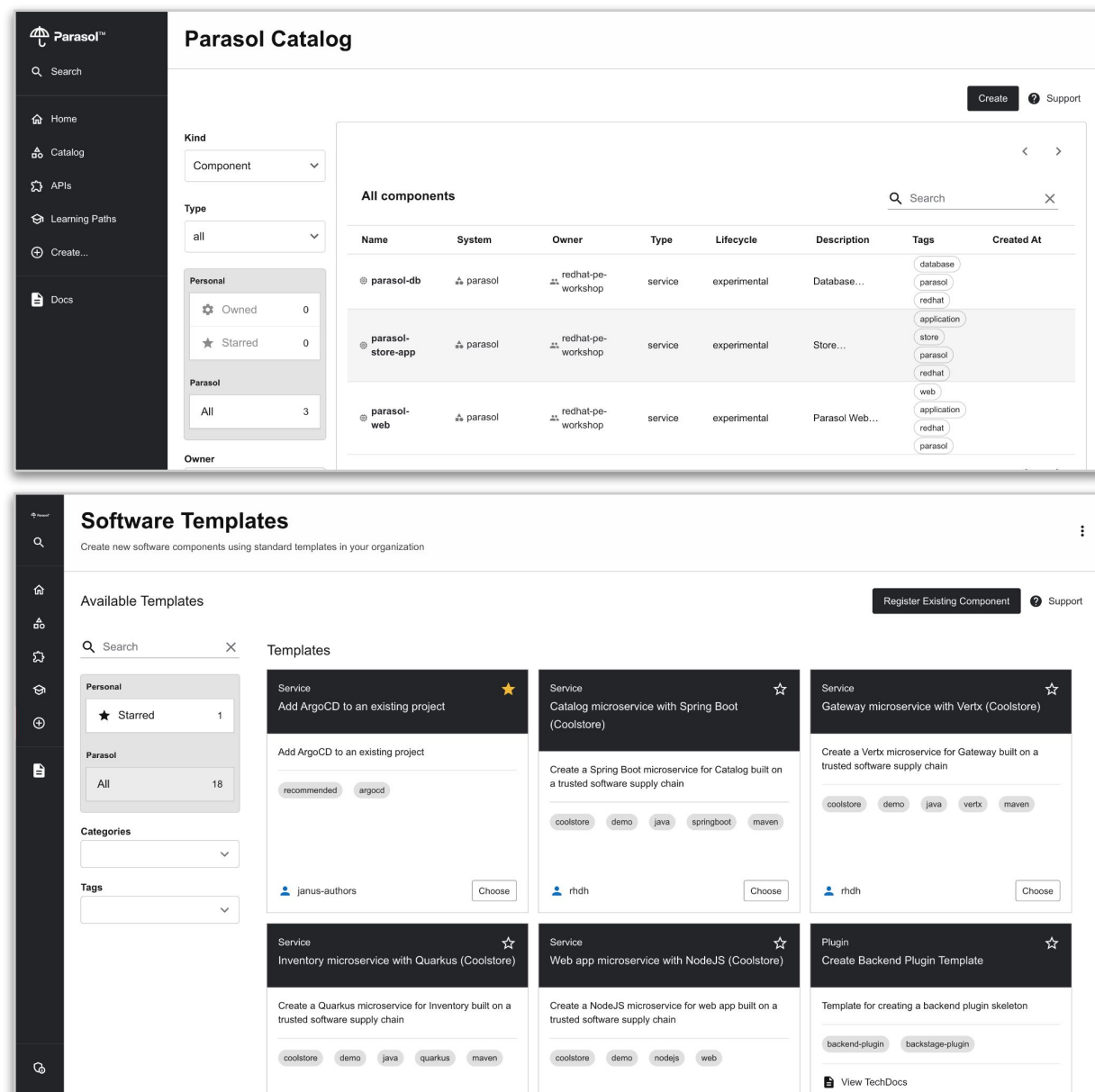
Red Hat OpenShift  
Dev Spaces  
Migration Toolkits

Products and related technologies that allow developers to **quickly and securely deliver applications to production**

Enabling platform engineers to more easily enforce operational controls and compliance across hybrid multi-cloud environments



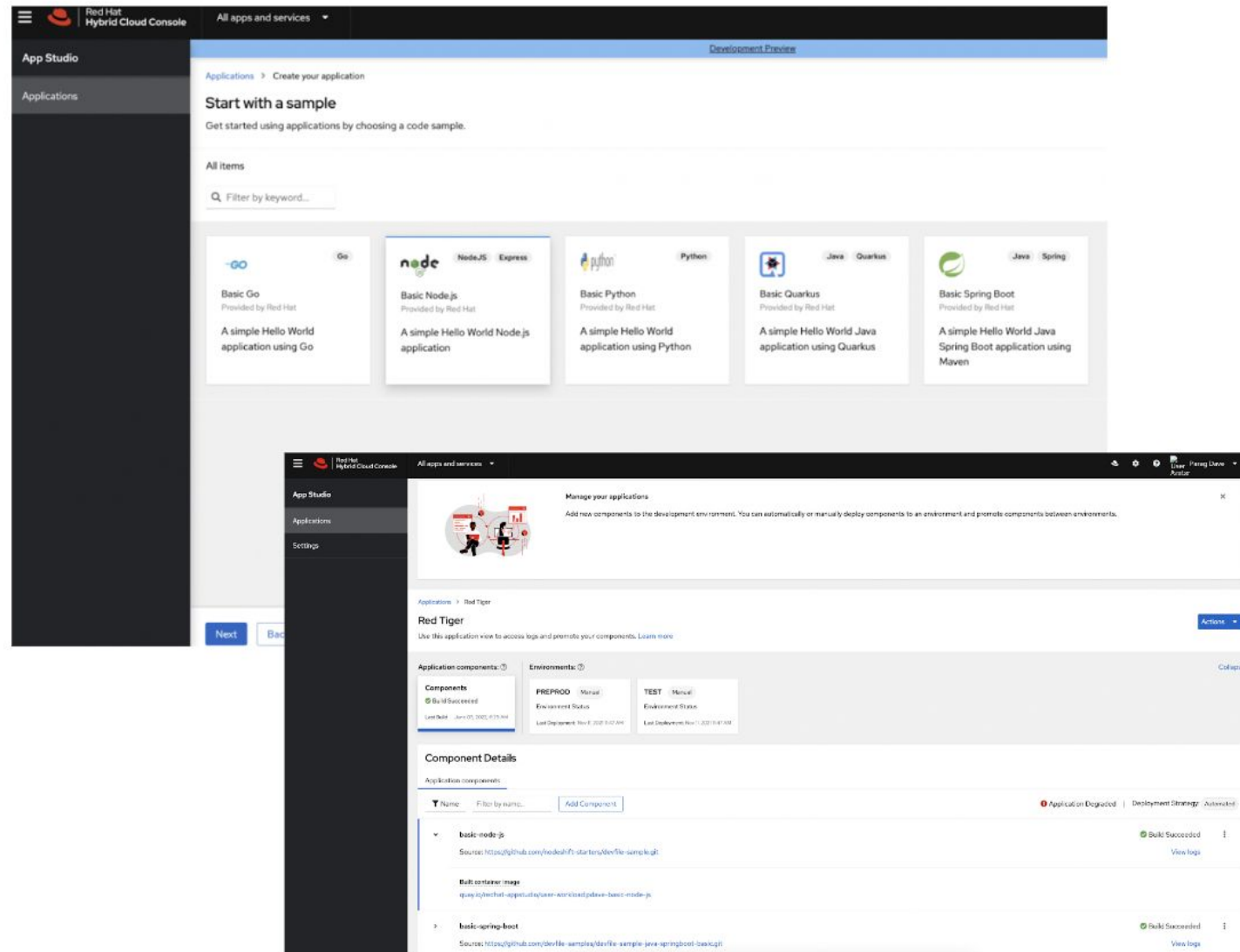
# Key Technical Capabilities: Developer Productivity



- Accelerated onboarding and frictionless self-service
- Single-pane visibility to simplify app ownership and governance
- Consistent Development-to-Production Workflows
- Automated Software Templates
- Streamlined Automation for Faster, Secure Development
- Rootless, Secure Container Execution



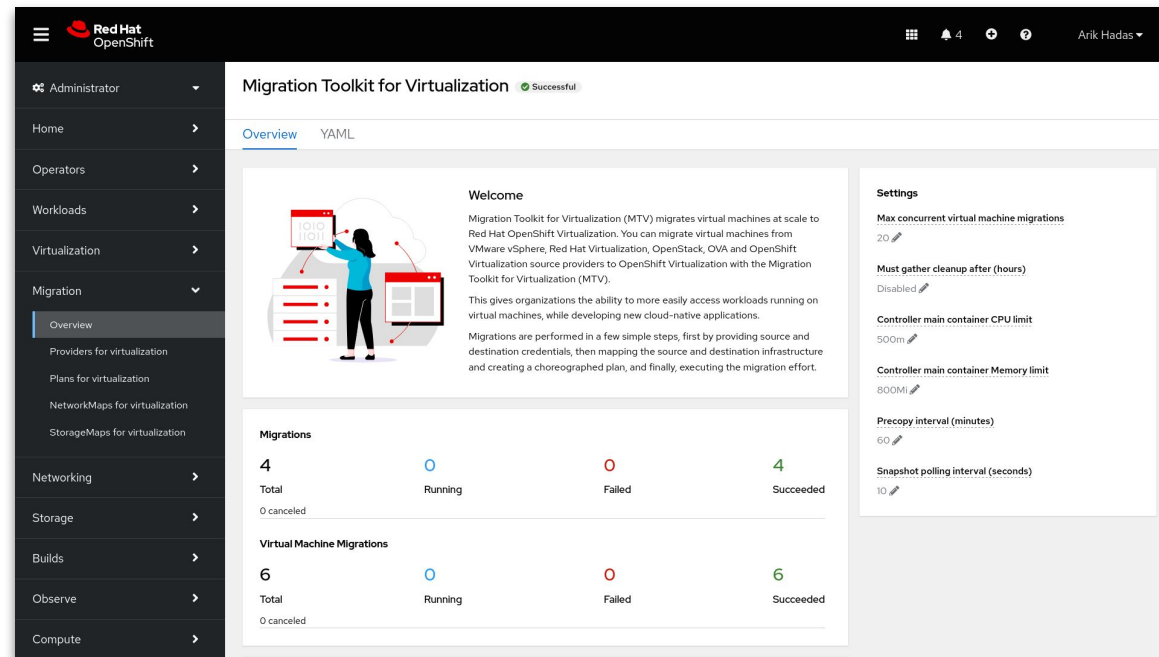
# Key Technical Capabilities: Supply Chain Security



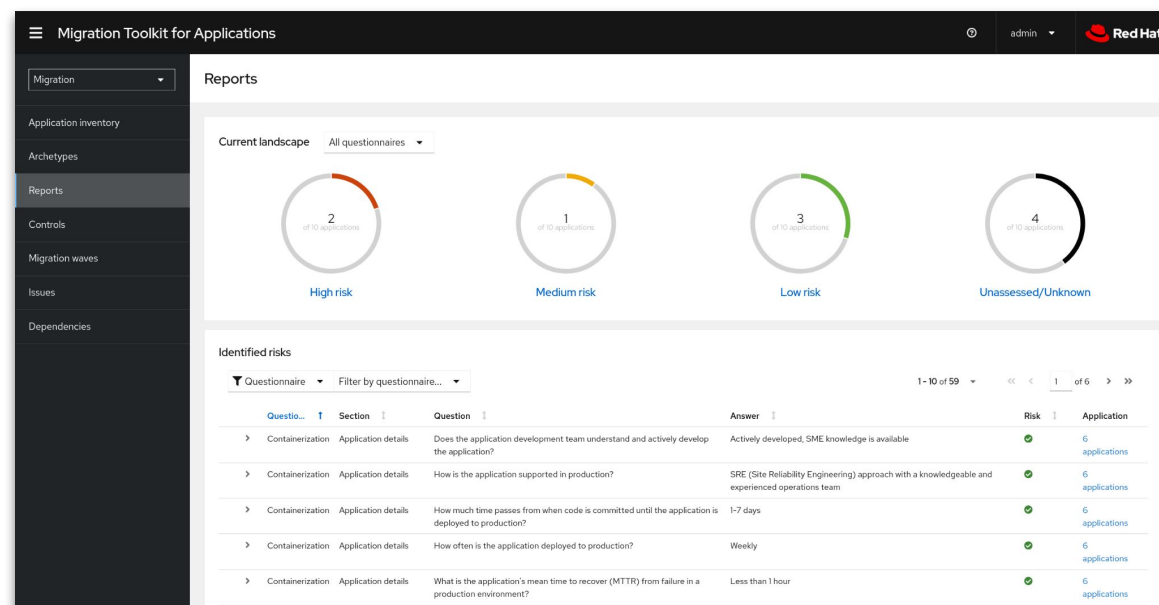
- Early artifact signing, verification, and secure builds within CI/CD system
- Automated signing & verification, provenance & attestations with immutable ledger
- Trusted Content and Policy-Based Remediation Guidance
- Risk Profile Monitoring & Blast Radius Analysis
- Software Composition Analysis (SCA)
- Software Bill of Materials (SBOMs) Generation & Management
- SLSA Framework – Secure software build pipelines with provenance guarantees
- Integration with ACS for buildtime policy checks, network policy generation, live application scanning, cluster-level security policy enforcement, automated compliance and deep insights



# Key Technical Capabilities: App Infra Modernization



- ☐ Integration with CI/CD Pipelines
- ☐ Automated Workload Migration
- ☐ AI-assisted Workload Modernization
- ☐ Docker-Compatible, Rootless Container Management





# Expanded OpenShift capabilities across the lifecycle

## OPENSIFT FOUNDATIONAL CAPABILITIES

## ADVANCED DEVELOPER SUITE CAPABILITIES

### CODE

Access cluster resources via web console  
IDE/desktop tools for local development  
Customizable CDE for remote development

Single point of entry portal for development teams  
Early risk scanning for vulnerabilities during coding  
Fast onboarding via self-service  
templates/docs/tools

### BUILD

Core CI/CD via OpenShift Pipelines and GitOps  
Integrated image registry  
Automated security and compliance in build

Automated signing/verification/tracing of code changes  
Automated supply chain integrity and SLSA enforcement  
Guidance, templates, tools to catch misconfigurations

### RELEASE

Storing, managing, and versioning application images  
Automated promotion of releases via GitOps  
Change tracking and rollbacks

Artifact signing (containers & AI models)  
Generate provenance, attestation, and SBOM metadata  
Complements ACS for supply chain integrity

### DEPLOY

Variety of deployment strategies (blue/green, etc)  
GitOps enables declarative continuous delivery  
Tagging and rollback, automated triggers

Application validation (signature/attestation/policy)  
AI model validations (signature/attestation)

### MONITOR

Real-time monitoring of cluster and app performance  
Centralized logging  
Alerting and health dashboards

Continuous Risk Monitoring & Blast Radius Impact  
Vulnerabilities & License Reporting

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